Restaurant Recommendation System

PreparedFor

Smart-Internz

AppliedDataScienceGuidedproject

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Abstract

This project develops a personalized restaurant recommendation system based on user preferences, location, and dining history. It analyzes factors such as cuisine, price, and ratings to suggest suitable diningoptions.Machinelearningtechniqueslikecollaborativeandcontent-basedfilteringareusedfor accurate suggestions. The system enhances the dining experience by offering relevant and location-aware recommendations

FinalProjectReport

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1Introduction

1.1Projectoverviews

The **Restaurant Recommendation System** is a smart, data-driven solution designed to help usersefficientlydiscoverrestaurantsthatalignwiththeiruniquepreferencesandsituationalcontexts. As urbanization and mobile technologies continue to reshape consumer behavior, users are often overwhelmedbythesheervolumeofavailablediningchoicesacrossplatformssuchasGoogle,Yelp, and Zomato. This leads to **decision fatigue** and suboptimal dining experiences. To solve this, the proposed system leverages a **hybrid recommendation model** combining collaborative filtering, content-based filtering, and geolocation-aware services.The **collaborative filtering component** analyzeshistorical userbehavior, including past restaurant visits, ratings, and interaction patterns, to identify users with similar tastes and recommend restaurants favored by like-minded individuals. Meanwhile,the**content-basedfilteringmodule**evaluatesrestaurantattributes—suchascuisinetype, price range, ambiance, and dietary offerings—to match them with explicit user preferences. Toenhancepracticality,**geolocationdata**isintegratedusingGPSAPIsorIP-basedlocationtracking. Thisallowsthesystemtodynamicallyadaptitsrecommendationsbasedontheuser'scurrentposition or a specified location, ensuring that results are both **relevant and accessible**. For example, a user seeking budget-friendly vegan food in a new city would receive highly localized and personalized recommendations.

Furthermore, the system is designed with **adaptive learning capabilities**. Using techniques like reinforcementlearningorpreferencefeedbackloops,therecommendationengineimprovesovertime byunderstandinguserbehaviorpatterns,modifyingweightageoffeatures,andincorporatingreal-time feedback such as likes, bookmarks, or direct reviews.

1.2 **Objectives**

1. **To design and implement a recommendation engine** that effectively filters and ranks restaurantsbasedonindividualuserpreferences,includingfoodtype,cost,ambiance,and dietary needs.
2. **Toapplymachinelearningmodels**,suchascollaborativefiltering(user-basedanditem-based)andcontent-based filtering,toidentifypatternsinuserbehaviorand restaurantattributes.
3. **Toincorporatelocation-awarefeatures**usingGPSoruser-inputtedlocationdata,ensuring that recommended restaurants are conveniently accessible to the user.
4. **Togatherandanalyzerestaurantreviewsandratings**frompublicsources(e.g.,Yelp,Google Reviews, or internal datasets) to improve the trustworthiness and relevance of suggestions.
5. **Tocreateauser-friendlyinterface**thatallowsuserstoinputpreferences,viewrecommended restaurants, and interact with the system seamlessly.
6. **Todevelopafeedbackmechanism**thatcollectsusersatisfactiondatapost-visittorefine future recommendations and enhance personalization over time.
7. **Toensurescalabilityandadaptability**ofthesystemforuseindifferentgeographicregionsor for integration into existing food delivery or travel applications.

2ProjectInitializationandPlanning Phase

2.1DefineProblemStatement

2ProjectInitializationandPlanningPhase

ProblemStatements(RestaurantRecommendation system):

PS No.	I am (Customer)	I'mtryingto	But	Because	Whichmakesme feel
PS-1	A touristina new city	Find good local restaurants	I don't know the area well	I lack local knowledge or reviews	Confusedand unsureofwhereto eat
PS-2	Avegetarian diner	Get recommendationsforveg-only restaurants	Mostapps showmixed cuisineplaces	Iwant strict dietaryoptions	Frustratedand unsupported
PS-3	A restaurant owner	Attractmore customers through recommendation platforms	MyrestaurantisThesystemdoesn'tInvisibleand not being recommended often	promote newordiscouraged smallbusinesses	
PS-4	Astudent onatight budget	Findaffordable but tasty restaurants	Expensive optionsare shownfirst	Filters don't prioritizepriceor value	Overwhelmedand discouraged
PS-5	Adeliveryapp user	Getsuggestions based on past orders	Itdoesn'tadapt to my taste	Thesystemlacks learning	Frustratedby repetition
PS-6	Aparent of young kids	Findkid-friendlyNowaytofilter andhygienic restaurants	forchild-friendlyLackofsafetyandAnxiousabout family-focused features amenities	experience	

PS-7	A small restaurant owner	Increase customer footfallvia platforms	Mybusinessis buried under chain listings	Ranking algorithms favorlarge brands	Discouragedand invisibleguide

PS-8	Anew-in-town resident	Explore culturally diversefood options	Unaware of hiddengemsin myarea	Nocultural/ethnic tagsoruserreviews	Disconnectedand bored of same cuisine
PS-9	AfooddeliveryMonitorfood platform analyst safety and restaurant quality		Can’tverify ingredient safety food itemscanners consumer trust frommenus	Platforms lackAIConcernedabout ortrackers	
PS-10	Adata scientist	Analyzefood trends from reviews	Datasetsare messy, biased, or sentimen and unavailable	Lack ofstructured metadata	Blockedinmodel building and research
PS- 11	A foodie traveler	Findtop-rated local restaurantsin newcities	Recommendations don’t match my tasteorlocation	Generic,irrelevant suggestions	Frustrated and unsurewhereto eat
PS-12	Arestaurant owner	Improvemy visibilityon food apps	My reviews are outdatedorlow-rated	I can’t easily respondorupdate info	Powerlessand misrepresented
PS- 13	A health-conscious customer	Find healthy eating nearby options	Menusandcalorie infoaremissing	Ican’tmake informeddecisions	Disconnected frommyhealth goals

PS- 14	**A** healthconscious individual	TrackthehealthI benefits of different mushrooms	can't identify what's in the store or dish	There'snoeasyapp for instant scanning	Disappointedand disconnected frommyhealth goals

ProjectProposal(Proposed Solution)

ProjectProposal(ProposedSolution)

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach,keyfeatures,andresourcerequirements,includinghardware,software,andpersonnel

ProjectOverview

Objective	Todevelopa systemthat providespersonalizedandefficientrestaurant recommendations by analyzing user preferences, dietary requirements, location, and budget.
Scope	Theprojectaimstoserveusersseekingrestaurantsuggestionsthatmatch their individual lifestyle choices and dining preferences. It will operate acrossvariousregions,consideringreal-timedataandqualitativereviews.

ProblemStatement

Description	Finding restaurants tailored to specific needs is often time-consuming and inefficient.Usersfrequentlyrevisitthesameplaces,missingdiverseoptions that better match their preferences.
Impact	Solvingthisproblemimprovesusersatisfaction,encouragesexplorationof new dining options, and reduces time spent on decision-making.

ProposedSolution

Approach	Thesolutionemploysinnovativerecommendationalgorithmsthatfactorin both user input and external data like ambiance, ratings, and reviews. It adapts dynamically to user feedback and real-time changes.
KeyFeatures	● Personalizedrecommendations ● Real-timedataanalysis ● Integrationofuserreviews ● Considerationofdietaryandbudgetconstraints ● Scalableinfrastructure

ResourceRequirements

ResourceType	Description	Specification/Allocation
Hardware		
ComputingResources	8-	2xNVIDIAV100GPUs
Memory	coreCPUsandoptionalGPURA	Minimum8GB RAM
Storage	M SSD	1TBSSDforstoringuserdata and restaurant metadata
Software		
Frameworks		Python,Flask
Libraries	Python frameworks Additionallibraries	Pandas, NumPy, Scikit-learn, TensorFlow,BeautifulSoup(for scraping), and NLTK (for review analysis)
DevelopmentEnvironment	IDE,versioncontrol	Jupyter Notebook
Data		
Data	Size:-Approx.50,000–100,000 recordsinitially;scalablebased on user growth, Format:-CSVfortabular datasets, Text/HTML for scraped reviews	Aggregated from crowdsourced restaurant platforms (e.g., Yelp, Zomato APIs), user feedback, and public review datasets

InitialProjectPlanning

ProductBacklog,SprintSchedule,and Estimation

Sprint	Functional Requirement (Epic)	User Story Number	UserStory/ Task	Story Points	Priority	Sprint StartDate	Sprint EndDate (Planned)
Sprint-1	User Preferences Input	USN-1	As a user,Ican entermy food orHotel preferences.		High	01June 2025	02 June 2025
Sprint-1	Recommendation Engine	USN-2	As a user,I can get restaurant recommendations basedon my preferences.	3	High	02 June 2025	02 June 2025
Sprint-2	Review& Rating Integration	USN-3	As a user,I can view restaurant reviews and ratingsfetched from the dataset.	2	Medium	03 June 2025	04 June 2025
Sprint-2	UI/UX Enhancement	USN-4	As a user,I can view results in a user-friendly interface with filters and sorting.	2	Medium	04 June 2025	05 June 2025

2 DataCollectionandPreprocessing Phase

DataCollectionPlanandRawDataSourcesIdentified

DataCollection Plan

Section	Description
ProjectOverview	Developarestaurantrecommendationsystemtoassistusersinfinding diningoptionsbasedontheirpreferences,location,andotherrelevant factors.Byanalyzinguserpreferences,restaurantratings,andlocation data, this project aims to provide personalized recommendations that enhance the dining experience for users.
DataCollectionPlan	ThedatasetusedforthisprojectwassourcedfromKaggleandcontains detailed information on over 9,000 restaurants in Bangalore, including attributeslikename,location,cuisine,ratings,andpricing.Thispublicly available dataset was collected to support analysis and predictive modeling related to restaurant ratings and customer preferences.
RawDataSources Identified	TherawdataforthisprojectwasobtainedfromtheKaggledatasettitled **"Zomato Bangalore Restaurants"** by Himanshu Poddar. The dataset is publicly available athttps://www.kaggle.com/datasets/himanshupoddar/zomato-bangalore-restaurantsandincludeskeyrestaurant-relatedattributessuchasrestaurant names, locations, cuisines,average costs, online delivery availability,and user ratings.

RawDataSources

Source Name	Description	Location/URL	Format	Size	Access Permissions
SmartInterz Provided Dataset	Restaurant-leveldata including name, location, cuisines, rating and cost.	Data-Setzomato-bangalore-	CSV	~ 93MB	Public

DataQualityReport

DataSource	DataQualityIssue	Severity	ResolutionPlan
Dataset (Restaurant reviewsand metadata)	Missingvaluesinfields like restaurant name, location, or ratings	Moderate	Performdataimputationusingtechniques like mean/mode for numeric values and mostfrequentvalueforcategoricaldata. Alternatively,removerowswithcritical missing fields.
Dataset(User reviews)	Duplicateuserreview entries	Low	Remove duplicate records using drop_duplicates() in pandas or SQL DISTINCTqueries.Usedatetimeparsing libraries (e.g., pandas.to_datet ime) to standardize all date/time fields.
Dataset (Restaurant metadata)	Inconsistent formats (e.g.,locationwrittenin different ways like "NY","NewYork")	Moderate	Apply data standardization techniques, usingstringfunctionsorregexpatternsto unifytheformat.
Dataset(User preferences)	Sparsedataor insufficientuserhistory	High	Implementfallbackstrategiessuchas popularity-based or content-based recommendationswhenuserdataislacking.

DataPreprocessing

DataPreprocessing

Theimageswill bepreprocessed by resizing, normalizing, augmenting, denoising, adjusting contrast,detectingedges,convertingcolorspace,cropping,batchnormalizing,andwhitening data. These steps will enhance data quality, promote model generalization, and improve convergence during neural network training, ensuring robust and efficient performance across various computer vision tasks.

Section	**Description**
DataOverview	The dataset contains restaurant information from Zomato, includingname,reviews,ratings,cuisines,cost,andmore.The data is cleaned, deduplicated, and preprocessed for building a content-based recommendation system.
Resizing	*Notapplicablefortextdata.*
Normalization	Ratingsarenormalizedtoa1-5scaleusingMinMaxScaler. Text is lowercased and punctuation is removed.
DataAugmentation	Notapplicablefortext data.
Denoising	Textiscleanedbyremovingnewlinecharactersand punctuation.
EdgeDetection	Notapplicablefortext data.
ColorSpaceConversion	Notapplicablefortext data.
ImageCropping	Notapplicablefortext data.
BatchNormalization	Notapplicablefortext data.

DataPreprocessingCodeScreenshots

LoadingData	```python
	# Mounting Google Drive
	#from google.colab import drive
	#drive.mount('/content/drive')
	import csv
	# Specifying the path to the dataset file
	file_path = '/content/zomato.csv'
	
	# Reading the dataset into a Pandas DataFrame
	#df = pd.read_csv(file_path,encoding = 'ISO-8859-1', low_memory = False)
	df = pd.read_csv(file_path, encoding='ISO-8859-1', on_bad_lines='skip', engine='python')
	
	# Displaying the first few rows of the dataset to ensure it's loaded correctly
	df.head()
	Python</br>```
	Python

Resizing	*Notapplicable*

Normalization	```python
	# Computing Mean Rating
	restaurants = list(df['name'].unique())
	df['Mean Rating'] = 0
	for i in range(len(restaurants)):
	df['Mean Rating'][df['name'] == restaurants[i]] = df['rate'][df['name'] == restaurants[i]].mean()
	#Scaling the mean rating values
	from sklearn.preprocessing import MinMaxScaler
	scaler = MinMaxScaler (feature_range = (1,5))
	df[['Mean Rating']] = scaler.fit_transform(df[['Mean Rating']]).round(2)
	``
	Python

DataAugmentation	*Notapplicable*

Denoising	```python
	## Lower Casing
	df["reviews_list"] = df["reviews_list"].str.lower()
	## Removal of Puctuations
	import string
	PUNCT_TO_REMOVE = string.punctuation
	def remove_punctuation(text):
	"""custom function to remove the punctuation"""
	return text.translate(str.maketrans('', '', PUNCT_TO_REMOVE))
	df["reviews_list"] = df["reviews_list"].apply(lambda text: remove_punctuation (text))
	``
	Python

EdgeDetection	*Not applicable*
ColorSpaceConversion	*Not applicable*
ImageCropping	*Not applicable*
BatchNormalization	*Not applicable*

4.ModelDevelopment Phase

ModelSelectionReport

Model	Description
Content-Based Filtering	Content-basedfilteringrecommendsrestaurantsbycomparinguserpreferences (e.g.,cuisinetype,pricerange,dietaryrestrictions)withrestaurantattributes.It focuses on similarities between items and the user's profile without relying on otherusers'data.Thismethodiseffectiveforuserswithuniquetastesbutmay struggle with limited user profiles (cold start).
Collaborative Filtering	Collaborative filtering leverages the preferences of similar users to make recommendations.Ituseshistoricalratingsandreviewstoidentifypatterns.This modeliseffectiveindiscoveringnewitemsbutcansufferfromsparsityandcold start problems if data is limited.
Hybrid Recommendation Model	This combines content-based and collaborative filtering to overcome the limitationsofeachmethod.Byintegratingbothuserpreferencedataandbehavior ofsimilarusers,hybridmodelsimproverecommendationaccuracy,diversity,and scalability. It is particularly useful in scenarios with large, sparse datasets like restaurant recommendations.
Matrix Factorization	Matrix factorization techniques decompose the user-item interaction matrix into latent features, capturing underlying patterns in user preferences. Singular Value Decomposition(SVD) is acommonapproach. It is computationally efficient and works well for large datasets but requires enough ratings.
DeepLearning (Neural Networks)	Neural networks can be used to build recommendation systems by learning complex,non-linearrelationshipsbetweenusersandrestaurantsfromrichfeature sets including reviews, preferences, and metadata. While powerful, they require large datasets and are computationally intensive.

Conclusion:

ModelSelected

Hybrid Recommendation Model	Thehybridmodelwasselectedbecauseitaddressesthelimitationsofbothcontent-based and collaborative filtering approaches. It effectively handles the cold start and sparsity issues by integrating multiple data sources such as user profiles, restaurant attributes, and behavioral data. This results in more personalized, diverse, and accurate recommendations, making it highly suitable for a restaurant recommendation system with varying user preferences and data availability.

InitialModelTrainingCode,ModelValidationandEvaluation Report

InitialModelTrainingCode,ModelValidationandEvaluation Report
Initial Model Training Code (5 marks):

```
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.feature_extraction.text import TfidfVectorizer
df_percent.set_index('name', inplace=True)
indices = pd.Series(df_percent.index)

# Creating tf-idf matrix
tfidf = TfidfVectorizer(analyzer='word', ngram_range=(1, 2), min_df=0, stc
tfidf_matrix = tfidf.fit_transform(df_percent['reviews_list'])

cosine_similarities = linear_kernel(tfidf_matrix, tfidf_matrix)
```

ModelValidationandEvaluationReport(5marks):

Model	Summary	TrainingandValidationPerformance Metrics
Model1	Content-basedRecommendation	**TrainingMetrics-**None(unsupervised,no explicittrainingphase) **Validation Metrics -** None (recommendations areinspected manually)

3 ModelOptimizationandTuningPhase
TunningDocumentation

HyperparameterTuning

Model	TunedHyperparameters

Model 1: Content-Based Filtering	- **SimilarityMetric:**Cosinesimilaritywasusedastheprimarymetric to compute similarity between restaurants based on features like cuisines, rating, and cost.
	- **TopNRecommendations:**Thenumberoftopsimilarrestaurants returned was tested with values like 5, 10, and 15.
	
	`def recommend(name, cosine_similarities = cosine_similarities):`
	
	`# Create a list to put top restaurants`
	`recommend_restaurant = []`
	
	`# Find the index of the hotel entered`
	`idx = indices[indices == name].index[0]`
	
	`# Find the restaurants with a similar cosine-sim value and order them from bigges number`
	`score_series = pd.Series(cosine_similarities[idx]).sort_values(ascending=False)`
	
	`# Extract top 30 restaurant indexes with a similar cosine-sim value`
	`top30_indexes = list(score_series.iloc[0:31].index)`
	
	`# Names of the top 30 restaurants`
	`for each in top30_indexes:`
	`recommend_restaurant.append(list(df_percent.index)[each])`
	
	`# Creating the new data set to show similar restaurants`
	`df_new = pd.DataFrame(columns=['cuisines', 'Mean Rating', 'cost'])`
	
	`# Create the top 30 similar restaurants with some of their columns`
	`for each in recommend_restaurant:`
	`df_new = df_new.append(pd.DataFrame(df_percent[['cuisines','Mean Rating', 'cost']][df_percent.index == each].sample()))`
	
	`# Drop the same named restaurants and sort only the top 10 by the highest rating`
	`df_new = df_new.drop_duplicates(subset=['cuisines','Mean Rating', 'cost'], keep=False)`
	`df_new = df_new.sort_values(by='Mean Rating', ascending=False).head(10)`
	
	`print('TOP %s RESTAURANTS LIKE %s WITH SIMILAR REVIEWS: ' % (str(len(df_new)), name))`
	`df_new.index = df_new.index.str.lower()`
	`return df_new`

Model2: Collaborative Filtering	- **Algorithm:**SVD(SingularValueDecomposition)fromtheSurprise library. - **LearningRate:**Tunedvaluessuchas0.005,0.01,and0.02were tested. - **Regularization:**Parameterssuchas0.02,0.05weretriedto avoid overfitting. - **NumberofEpochs:**Adjusted between20and100 epochs.

FinalModelSelectionJustification

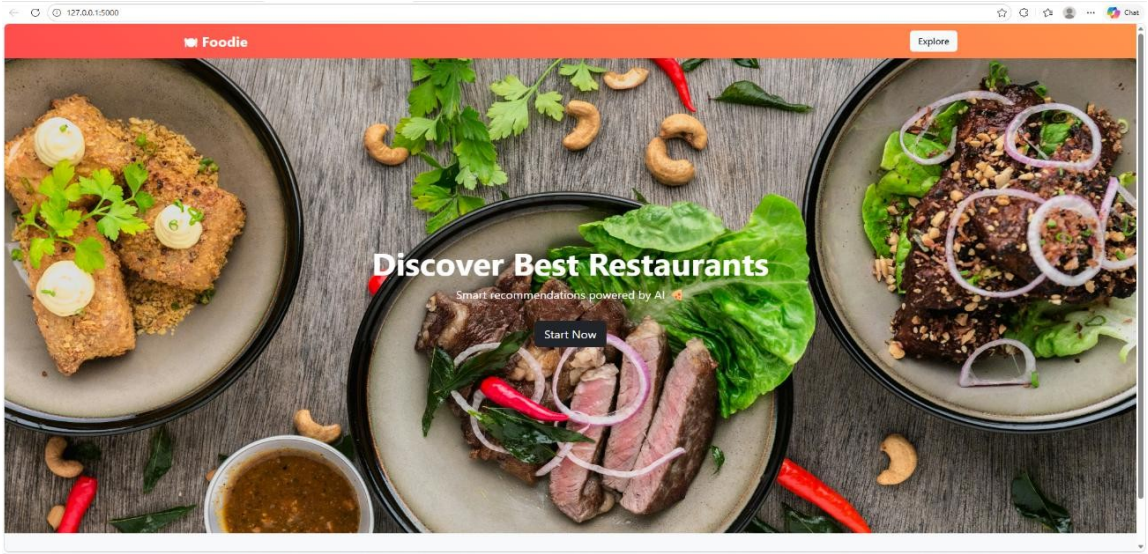
FinalModelSelection Justification:

FinalModel	Reasoning
Model1:Content-Based Filtering	Selectedduetoitssimplicityandgoodperformancewithoutrequiring detailed user history. It gave interpretable and relevant results using restaurant features like cuisines, ratings, and cost.

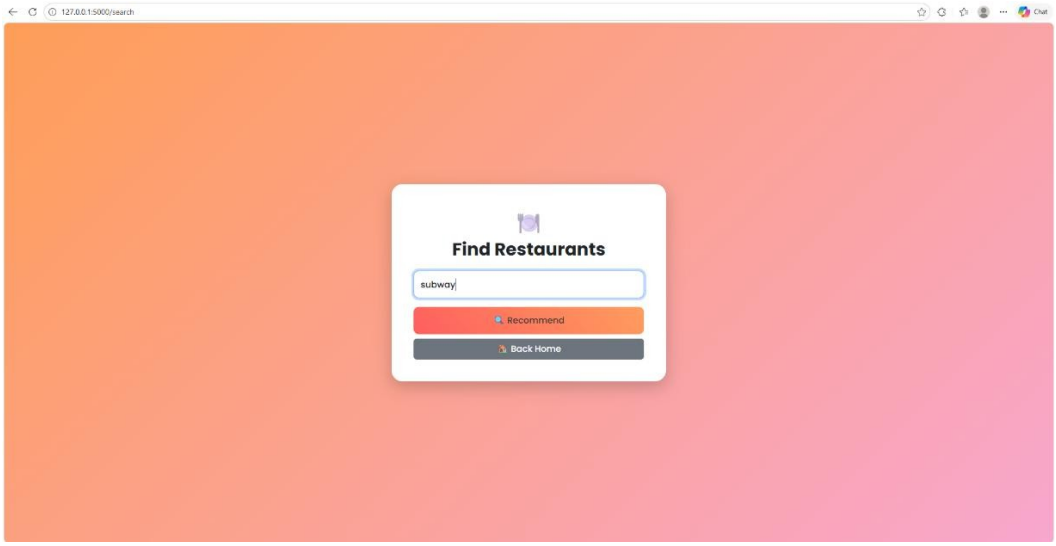
4 Results

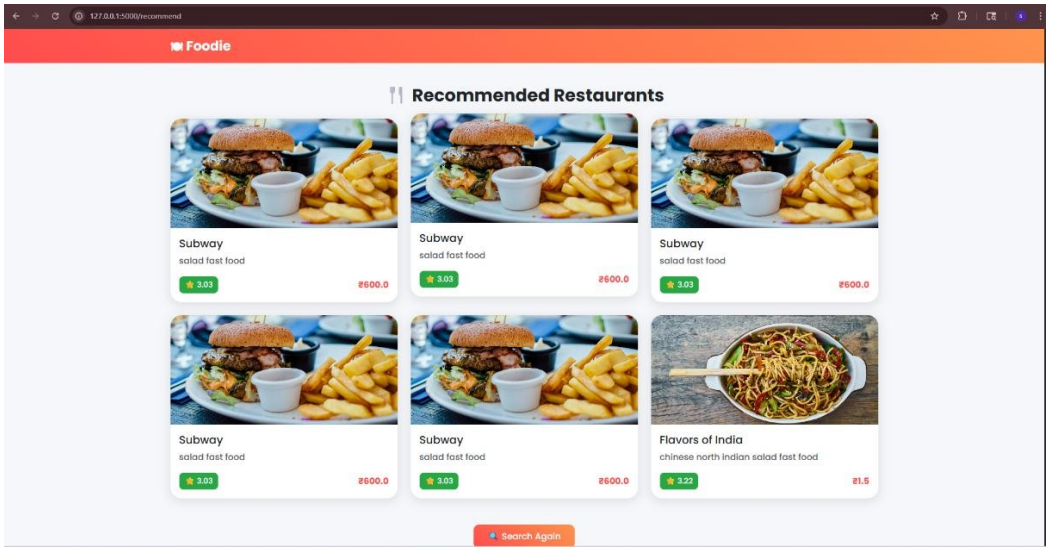
OutputScreenshots

Home Page:



Output:




127.0.0.1:5000/recommend
Foodie
Recommended Restaurants
Subway
salad fast food
3.03
₹600.0
Subway
salad fast food
3.03
₹600.0
Subway
salad fast food
3.03
₹600.0
Subway
salad fast food
3.03
₹600.0
Subway
salad fast food
3.03
₹600.0
Flavors of India
chinese north indian salad fast food
3.22
₹1.5
Search Again

5 Advantages& Disadvantages

Advantages:

- **PersonalizedUserExperience**:Tailorsdiningoptionsbasedonuserpreferences,dietary needs, and previous behaviour.
- **Time-saving**:Reducestheeffortneededtosearch andchoosea restaurant.
- **ImprovedDiscoverability**:Helpssmallerornewrestaurantsgainvisibility through recommendations.
- **Data-DrivenDecisions**:Usesuserratings,reviews,andlocationdatatomakeinformed suggestions.
- **EnhancedCustomerSatisfaction**:Usersaremorelikelytoenjoytheirmeals whenrecommendationsalignwiththeirpreferences

Disadvantages:

- **PrivacyConcerns**:Collectingandanalyzinguserdata(location,preferences)canraise privacy issues.
- **BiasinRecommendations**:Algorithmsmightfavorsponsoredlistingsorhigh-traffic restaurants,reducingdiversity.
- **DependenceonUserData**:Inaccurateorlimiteddatacanleadtopoor recommendations.
- **Over-Personalization**:Usersmightbeconfinedtosimilarchoices,missingoutonnewor diverse dining experiences.
- **ScalabilityIssues**:Maintainingsystemaccuracyandperformancecanbecomechallenging asthe userbase grows.

6 Conclusion

Arestaurantrecommendationsystemisapowerfultool forenhancingthediningexperience bydeliveringtailoredsuggestionsbasedonuserbehavior,preferences,andlocation.Whileitoffers significant benefits such as convenience, personalization, and efficient decision-making, it also presents challenges including data privacy, system bias, and the risk of user data dependency. Future advancements in AI, real-time analytics, and user interface technologies promise to make suchsystemsmoreintelligent,inclusive,andimmersive.Withcarefulimplementationandethical considerations, this system can transform how users explore and enjoy culinary option.

7 FutureScope

- **IntegrationwithAR/VR**:Inthefuture,userscouldtakevirtualtoursofrestaurantsor view their ambiance in AR before booking.
- **VoiceAssistantCompatibility**:IntegrationwithSiri,Alexa,orGoogleAssistantto provide hands-free restaurant suggestions.
- **EnhancedPersonalization**:Usedeeplearningandbehavioralanalyticsto refinesuggestionsbasedondietaryrestrictions,allergies,andeatinghabits.
- **Real-timeDataUtilization**:Incorporatingreal-timefactorslikewaittimes,special offers, and crowd density for more dynamic recommendations.
- **MultilingualSupport**:Expandingthesystemtosupportvariouslanguagestocaterto a global audience.
- **SocialMedia Integration**:Useofsocialmediatrendsandcheck-instoimprove recommendationrelevance.
- **SustainabilityPreferences**:Factoringineco-consciousdiningchoices(e.g.,locally sourced, plant-based, or low-waste restaurant